

# ALHAD BHADDEKAR

alhad.bhadekar@gmail.com | [linkedin.com/in/alhadb](https://www.linkedin.com/in/alhadb) | [alhadbhadekar.github.io/my-portfolio](https://alhadbhadekar.github.io/my-portfolio) | (215) 397-8395

## SUMMARY

- Senior Software Engineer and AI/ML Engineer with 6+ years of software development experience building distributed backend systems, cloud-native platforms, APIs, and production AI/ML applications.
- Strong experience designing and developing scalable systems using Python, Go, FastAPI, REST, GraphQL, gRPC, AWS, Kubernetes, Redis, relational/NoSQL databases, and event-driven architectures.
- Hands-on experience building production machine learning and Generative AI systems using LLMs, Retrieval-Augmented Generation (RAG), AI agents, semantic/vector search, anomaly detection, PyTorch, Scikit-learn, Transformers, LangChain, and LangGraph.
- Experienced in production engineering, system performance optimization, observability, CI/CD, infrastructure automation, and troubleshooting distributed systems at scale.

## TECHNICAL SKILLS

**AI/ML:** Machine Learning, Deep Learning, NLP, Scikit-learn, PyTorch, TensorFlow/Keras, Transformers, SBERT, BERT, RoBERTa, T5, XGBoost, Random Forest, Isolation Forest

**Generative AI & LLMs:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, Hugging Face, OpenAI APIs, Prompt Engineering, Tool Calling

**Search & Retrieval:** Embeddings, Semantic Search, Vector Search, FAISS, Chroma, Elasticsearch

**Data & ML Engineering:** Python, NumPy, Pandas, model deployment, inference optimization, automated retraining, model evaluation, MLOps

**Backend Engineering:** Python, Go (Golang), FastAPI, Django, Flask, Gin, REST, GraphQL, gRPC, Protobuf, Microservices, Event-Driven Architecture

**Cloud & MLOps:** AWS (EKS, ECS, Lambda, S3, SQS, SNS, RDS/Aurora), Kubernetes, Docker, Terraform, Helm, CI/CD

**Databases & Observability:** PostgreSQL, MySQL, MSSQL, MongoDB, Redis, DynamoDB, Cassandra; Prometheus, VictoriaMetrics, Grafana, ELK Stack, CloudWatch

## EXPERIENCE

### Comcast

#### Software Development Engineer 3 | Feb 2023 - Present

- Built distributed backend services in **Python and Go across 10+ services** supporting large-scale network inventory, topology analysis, and AI-driven operational workflows.
- Developed **NomadDB**, a cloud-native platform exposing **REST and GraphQL APIs** for network inventory, connectivity, and topology data consumed by downstream services and AI workflows.
- Contributed to **Graph Smasher**, a graph-based topology analysis system using **Python and NetworkX** for dependency analysis, conflict detection, connectivity analysis, and network isolation.
- Built secure **REST, GraphQL, and gRPC/Protobuf APIs** with JWT authentication and role-based access control for communication across distributed backend services.
- Built production observability using **Prometheus, VictoriaMetrics, Grafana, and ELK**, improving service monitoring, incident detection, and production troubleshooting.
- Built CI/CD and infrastructure automation using **Jenkins, Concourse, Docker, Kubernetes, Terraform, Helm, and AWS** for repeatable deployment of AI and backend services.
- Built and deployed a hybrid **machine-learning anomaly detection platform** using **Isolation Forest and SBERT embeddings** to identify infrastructure anomalies and reduce noisy operational alerts.
- Developed production **Retrieval-Augmented Generation (RAG) systems** using historical root-cause analysis data, operational runbooks, network inventory, and topology data to accelerate incident investigation and troubleshooting.
- Built **LLM-powered AI agents** integrating with REST and GraphQL APIs to retrieve network inventory, topology, and device data for validation, investigation, and remediation workflows.
- Developed **multi-agent workflows with LangChain and LangGraph**, implementing tool calling, state management, structured execution, and coordination across backend services.
- Built **embedding, semantic-search, and vector-retrieval pipelines** using **FAISS, Chroma, Elasticsearch, Hugging Face, and OpenAI embeddings** for contextual retrieval and real-time alert enrichment.
- Integrated **Prometheus/VictoriaMetrics metrics, logs, traces, and operational topology data** into LLM workflows for root-cause analysis and alert investigation.
- Implemented **LLM guardrails, structured tool execution, validation, and inference auditing** to improve reliability, security, and traceability of AI-agent workflows.
- Optimized LLM inference using **batching, caching, embedding optimization, and quantized Transformer models** to improve response latency and throughput under peak workloads.
- Built **Streamlit dashboards, LangServe APIs, and AWS deployment pipelines** for AI/ML model evaluation, monitoring, deployment, and operational visibility.

### Pelmorex Corp. - The Weather Network

#### Software Developer | Nov 2021 - Feb 2023

- Built high-performance **Go and Python microservices** handling **1,000+ requests per second with sub-100 ms latency** for data-driven weather applications.

- Developed REST, GraphQL, and **gRPC/Protobuf services** and implemented concurrent processing using Go goroutines and channels.
- Built cloud-native, event-driven systems on **AWS** using Lambda, ECS, S3, SQS, and SNS and helped migrate legacy components to scalable services.
- Worked with **geospatial PostgreSQL data** for location-aware weather applications and deployed Redis Sentinel as a distributed caching layer.
- Built automated testing and deployment pipelines using Jenkins, Docker, Kubernetes, Terraform, CloudFormation, and AWS; implemented monitoring with ELK, Grafana, and CloudWatch.
- Developed a production **AI-powered weather alert summarization system using OpenAI GPT**, transforming complex weather information into concise user-facing natural-language summaries.

## **The Habitat Learn**

**Backend Developer** | *Aug 2020 - Nov 2021*

- Built production backend services and REST/GraphQL APIs using **Python, Django, and Node.js** and developed data-driven full-stack functionality using React and Redux.
- Improved system and database performance through architecture and query optimization; deployed and monitored applications on **AWS** using Terraform, EC2, RDS, Route 53, and CloudWatch.
- Built a real-time IoT camera-management platform using **MQTT** for remote device communication and automation.

## **Royal Bank of Canada**

**Senior Network Security Analyst / Python Developer** | *Feb 2019 - Jul 2020*

- Built Python applications and backend services using **Django, Flask, and FastAPI** for automation and network-security workflows.
- Developed REST APIs and automated ingestion and processing of structured **JSON/XML operational data**; designed MSSQL data models and Python-based ingestion workflows.
- Built Python automation for security infrastructure, incidents, change data, reporting, and operational analysis; implemented testing with pytest and unittest.

## **Earlier Engineering Experience**

Secure Sense Solutions - Tier 2 Network Analyst (Feb 2018 - Feb 2019) | Forsythe Technology - Analyst II (May 2017 - Jan 2018) | Trustwave - TAC Analyst (Aug 2016 - May 2017) | Check Point Software Technologies - Tier-2 Network Security Support Engineer (Jan 2016 - Aug 2016)

Enterprise network/security engineering, production troubleshooting, SIEM, incident response, firewall technologies, ServiceNow, Bash automation, root-cause analysis, and infrastructure operations.

## **EDUCATION**

**Master of Electrical and Computer Engineering** - GPA: 4.15/4.3 | Concordia University, Montreal, Canada | May 2015

**Bachelor of Engineering, Electronics and Telecommunication Engineering** | University of Mumbai, India | Jun 2013